Sushant Payal

mail@sushantp.me | Pune, MH | GitHub: sushant-demn

EDUCATION

International Institute Of Information Technology, SPPU 2023 – Present
Bachelor of Engineering, Information Technology GPA: 7.8

EXPERIENCE

Java Full Stack Virtual Intern, AICTE – EduSkills (EduSkills Academy) Jan 2026 – Mar 2026

- Completed a 10-week Java Full Stack virtual internship covering Core and Advanced Java, Spring Framework, Spring Boot, Hibernate, MySQL and Git.
- Built an **API Rate Simulation System** in Java: a modular pipeline that parses CSV request logs, tracks per-user request counts using HashMaps, validates 60-second time windows, enforces per-user request limits (ALLOW/BLOCK) and reports blocked-request analytics.
- Defined clear input/output contracts (JSON output per stage) so each module's output feeds directly into the next.

PROJECTS

Pacman: The Hunt Protocol (Unity Engine, Photon PUN, C#) - Github

- Developed a 2D multiplayer game inspired by Pacman using Unity Engine and C#.
- Integrated Photon PUN for real-time multiplayer networking, lobby and player state synchronization.
- Developed a 2D grid-based movement system using BoxCast for precise collision detection and physics interactions.
- Built a dynamic, scalable level architecture using Unity Tilemap tools; implemented responsive UI and real-time spatial tracking indicators for player avatars.

Sustify – Hi-Res Audio Player (Miniaudio, WebView, TagLib, Portable File Dialogs, C++) - Github

- Developed a high-fidelity desktop application supporting lossless and compressed formats such as FLAC, WAV and MP3.
- Built a low-latency audio engine using the Miniaudio API for real-time state management, precise track seeking and continuous playback.
- Bridged native backend logic with a responsive web-based UI via WebView, using Portable File Dialogs for native cross-platform file selection.
- Integrated TagLib to extract and display embedded ID3 metadata (artist, album, track title) in real time.
- Architected a linear queueing data structure to manage track sequencing and ensure seamless audio transitions.

Residential Security System (C++, Arduino, Raspberry Pi, ESP32, MySQL, ReactJS, Node.js)

- Architected a dual-layer IoT system with a Raspberry Pi as central server and Arduino UNO for sensor data acquisition.
- Implemented a real-time alert subsystem that pushes instant security notifications via the Telegram API on threat detection.
- Engineered a live video surveillance feed using the Raspberry Pi Camera Module and old smartphones as remote cameras, accessible remotely through a web interface.
- Developed a full-stack web dashboard (Node.js, MySQL) to visualize real-time sensor data and manage security logs.

TECHNICAL SKILLS

Languages: C++, C#, Java, JavaScript, SQL
Frameworks & Libraries: ReactJS, Node.js, Photon PUN, TagLib, Miniaudio, WebView
Databases: MySQL
Tools & Hardware: Unity Engine, Arduino, Raspberry Pi, ESP32, Ubuntu Server, Git, GitHub, Microsoft Azure

ACHIEVEMENTS

Runner-Up, Game X Innovate: Hacksprint v7.0 (organised by MMCOE Karvenagar and Zensar Technologies)

- Developed a multiplayer 2D game in under 24 hours using Unity Engine and the Photon PUN framework.
- Secured Runner-Up position out of 50+ teams at national level.

CERTIFICATIONS

- C and C++ Training, Spoken Tutorial, IIT Bombay (EduPyramids, SWAYAM Plus), May 2025